

Chanbin Lim

Los Angeles, CA · +1 (213) 285-3564 · chanbin.lim.cs@gmail.com · linkedin.com/in/chanbin-lim/

SUMMARY

Software Engineer with 5 years of experience in performance optimization, reliability engineering, and automation across complex software systems. Skilled in debugging, trace/flow analysis, and building reproducible test and integration workflows. Currently pursuing an M.S. in Computer Science at USC.

WORK EXPERIENCE

Software Engineer

Samsung Electronics – MX Division (System Framework Team), Suwon, South Korea Sep 2020 – Jun 2025

- Optimized complex request flows across interdependent system components, reconstructing execution paths and isolating root causes for timeout, missing-response, and multi-service interaction issues.
- Redesigned managed runtime and compilation pipeline to achieve a 15% gain in application startup latency by restructuring execution paths and resource initialization to eliminate cold-start overhead.
- Eliminated reboot loops across 2+ Android OS releases spanning phones, wearables, and tablets, achieving over 95% recovery in matching cases by designing and integrating an autonomous runtime artifact restoration mechanism.
- Ensured platform compatibility across 5+ Android OS releases by continuously resolving dependency issues introduced through frequent upstream Google code updates.
- Delivered upstream platform fixes impacting millions of production devices by resolving 20+ late-stage, release-blocking platform issues in close collaboration with Google and Qualcomm engineers, drawing on experience from analyzing 1,000+ customer-reported issues to prioritize and reproduce critical failures.

Software Engineer

Continental Automotive Electronics, Sejong, South Korea Jan 2018 – Jul 2020

- Developed automated workflows to deploy build artifacts to devices and validated boot behavior using image-based analysis, reducing manual smoke-testing effort by 30% and increasing integration reliability during daily builds.
- Rebuilt and scaled an internal CAN stress test tool used by 200+ engineers across development organization, improving test stability and productivity by modernizing core libraries and refining user interface.
- Constructed reproducible multi-device test environments with CAN/Ethernet signal injection to execute stress tests, enabling 2+ times faster issue reproduction and earlier regression detection.

PATENTS (3 ISSUED, PRODUCTION-DEPLOYED)

- Reduced runtime-level application cold-start latency by 15% by introducing pre-execution profiling into the application startup path prior to first user-visible execution.
- Enabled autonomous recovery from recurring native reboot loops caused by corrupted system runtime artifacts, preventing factory resets through high-precision OS-level detection and safe artifact regeneration logic.
- Improved first-launch performance for frequently used applications by applying adaptive, usage-driven prioritization to system package compilation, preserving user-facing performance even under partial compilation conditions.

EDUCATION

University of Southern California

Master of Science in Computer Science, GPA : 3.85/4.00

Relevant Coursework: Algorithms, Database Systems, Web Technology, Machine Learning

Los Angeles, California, USA

Aug 2025 – May 2027

JEONBUK NATIONAL UNIVERSITY

Bachelor of Science in Electronics, GPA: 3.75/4.00

Jeonju, South Korea

Feb 2018

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript (React, TypeScript), HTML, CSS, Shell

Systems & Backend: Linux, REST APIs, Multithreaded Programming, OS-level Debugging, FastAPI, SQLite

Tooling & DevOps: Git, Docker, Docker Compose, Jenkins (CI/CD), JIRA, Gerrit

Testing & Analysis: Profiling (perf), Trace Analysis, Automation (Python/Shell)

Data & ML: PyTorch, Pandas, NumPy (basic), OpenAI API, Gemini API